Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination - APRIL 2021

CS204S – FUNDAMENTALS OF DATA STRUCTURES

Time : 3 hrs **Max Marks :75**

SECTION – A (5X5=25)

Answer any **FIVE** Questions

1. Explain the various operations on data structures.
2. Discuss the storage structure for arrays.
3. Write an algorithm to obtain the value of the ith element from the top of a stack without deleting it.
4. Draw a diagram of circular queue for its operations.
5. Write an algorithm for producing a postfix expression from an infix with an example.
6. Write an algorithm to insert a node in a linked list.
7. Write short notes on doubly linked list.
8. Define a graph and its types.

SECTION - B (5X10=50)

Answer any **FIVE** Questions

9. Explain the representation of linear arrays in memory.
10. Write a procedure to add and delete an element in a queue.

11. Design an algorithm to perform PUSH and POP operations on a stack.

12. Discuss about polynomial manipulation and multiple precision arithmetic.

13. Write the general algorithm for a preorder traversal of a binary tree using iteration.

14. Write an algorithm for inserting a node to the left of a given node in a doubly linked list.

15. Write a procedure to add two polynomials.

16. Explain Dijikstra's shortest path algorithm in detail.

****